

Bursa PalmAI User Guide

Welcome to **Bursa PalmAI**, the AI and Data Powered Palm Oil Intelligence Hub! This 5-page user guide helps you navigate the app's five main tabs—Market Overview, Company Profiles, Commodity Trends, Geography Insights, and Global Trade Chain—to access real-time insights on the palm oil industry. Each section below corresponds to one tab, explaining its features, how to interact with visualizations, and tips for optimal use. Bursa PalmAI is accessible at <https://bursa-palm.ai.onrender.com/> and loads the Market Overview tab by default.

Page 1: Market Overview Tab

Overview

The Market Overview tab provides a snapshot of the palm oil industry, featuring real-time metrics, interactive charts, and news updates. It's your starting point for understanding market sentiment, stock performance, and industry trends.

Features

- **News Sentiment Card:**
 - Shows the dominant sentiment (Positive, Neutral, Negative) of recent palm oil news.
 - Color-coded: Green (Positive), Yellow (Neutral), Red (Negative).
 - Displays the number of headlines analyzed (e.g., "Based on latest 10 headlines").
 - **How to Use:** View to gauge market perception. Updates automatically.
- **Price Momentum Card:**
 - Indicates the trend of palm oil company stock prices (Bullish, Bearish, Neutral).
 - Color-coded: Green (Bullish), Yellow (Neutral), Red (Bearish).
 - Based on recent share price changes.
 - **How to Use:** Check for market momentum. Updates automatically.
- **Weather Forecast Card:**
 - Shows the dominant 7-day weather forecast for palm oil estates (e.g., Cerah, Berangin, Hujan, Ribut Petir).
 - Color-coded: Green (Cerah), Yellow (Berangin), Orange (Hujan), Red (Ribut Petir).
 - **How to Use:** Assess weather impacts on production. Updates automatically.
- **Market Cap Treemap:**
 - Visualizes market capitalization of major palm oil companies (in RM billions).
 - Colors range from light to dark green based on size.
 - **How to Use:** Hover over a company to see its name and market cap (e.g., "IOI: RM 5.2 Billion").
- **KLCI Chart:**
 - A line chart of the Kuala Lumpur Composite Index (KLCI) over the last 7 days.

- Shows the latest value and percentage change (e.g., "+2.5%").
- **How to Use:** Hover to view daily values for market context.
- **Share Price Cards:**
 - Displays stock prices for companies (e.g., KLK, IOI) with logos, prices, and daily changes (e.g., "RM 4.50 ▲ 2.3%").
 - Green (up), Red (down), or Gray (no change) indicators.
 - Shows 4 companies per page.
 - **How to Use:** Click **Prev** or **Next** buttons to browse companies.
- **AI Summary:**
 - A concise AI-generated summary of industry trends (e.g., "Rising CPO prices drive revenue growth").
 - **How to Use:** Read for quick insights. Updates automatically.
- **Latest Palm Oil News:**
 - Lists up to 6 recent articles with headlines, descriptions, and dates.
 - **How to Use:** Click "Read more" to open the full article in a new tab.

Navigation Tips

- Access via the "Market Overview" link in the sidebar (with a chart-line icon).
- If data shows "Loading..." or "N/A," refresh the page or check your internet connection.
- All visuals are responsive and adjust to your screen size.

Page 2: Company Profiles Tab

Overview

The Company Profiles tab offers detailed insights into a selected palm oil company's performance, including production, stock prices, earnings, and forecasts.

Features

- **Company Selection:**
 - A dropdown menu lists companies (e.g., Kuala Lumpur Kepong Berhad, IOI Corporation Berhad).
 - **How to Use:** Select a company to update all visuals.
- **Company Logo and Description:**
 - Shows the company's logo and a brief overview of its operations.
 - **How to Use:** Read for context about the company.
- **Company Share Price Chart:**
 - A line chart of the company's stock closing prices (in RM) over time.
 - **How to Use:** Hover to see specific dates and prices.

- **Monthly Production Chart:**

- A bar chart of monthly production volumes (e.g., Fresh Fruit Bunches, Crude Palm Oil).
- **How to Use:** Hover over bars for exact volumes.

- **Company Quarterly Earnings Chart:**

- A bar chart comparing quarterly revenue and net profit (in RM billion), with a line for net profit margin (%).
- **How to Use:** Hover for quarterly values.

- **Planted Area Chart:**

- A doughnut chart showing planted area by category (e.g., mature, immature) in hectares for the latest year.
- **How to Use:** Hover for area values.

- **Extraction Rate Chart:**

- A line chart of annual extraction rates (%) by category (e.g., oil extraction rate).
- **How to Use:** Hover for yearly rates.

- **Revenue Forecast Chart:**

- A bar chart comparing the latest quarter's revenue to the next quarter's forecast (in RM million).
- Shows contributions from FFB, CPO, and PK as percentages (e.g., FFB: 70%, CPO: 20%, PK: 10%).
- Includes a warning if data is incomplete (e.g., "⚠ Inaccurate forecasts").
- **How to Use:** Review for revenue trends; note warnings for accuracy.

- **Correlation Heatmap:**

- A heatmap showing correlations (0 to 1) between revenue, CPO price, PK price, and FFB price.
- **How to Use:** Hover for correlation values (e.g., "Revenue vs CPO: 0.95").

- **Impact Analysis:**

- A dropdown (CPO, PK, FFB) shows AI-generated insights (e.g., "RM513 increase in CPO price leads to RM151 million revenue increase").
- **How to Use:** Select a metric to view its revenue impact.

Navigation Tips

- Access via the "Company Profiles" link in the sidebar (with a building icon).
- Visuals update instantly upon selecting a new company.
- Note the disclaimer: Forecasts are for informational purposes only, not financial advice.

Page 3: Commodity Trends Tab

Overview

The Commodity Trends tab tracks prices and trends for palm oil and related commodities, helping you monitor factors affecting plantation economics.

Features

- **MPOB Key Indicators Chart:**
 - A mixed chart with bars for MPOB data (e.g., production, stocks, exports) and a line for FFB price (in RM) by month.
 - **How to Use:** Hover for specific values (e.g., production volume).
- **CPO, FFB, and Palm Kernel Prices Chart:**
 - A line chart comparing Crude Palm Oil (CPO), Fresh Fruit Bunches (FFB), and Palm Kernel prices.
 - Left axis: CPO/PK prices (RM thousands); right axis: FFB price (RM tens).
 - **How to Use:** Hover to compare price trends.
- **Soybean Futures Chart:**
 - A line chart of soybean oil futures prices (in USD), a substitute for palm oil.
 - **How to Use:** Hover for daily prices to compare with palm oil.
- **Fertilizer Prices Chart:**
 - A line chart of fertilizer prices (e.g., urea, NPK) in RM, impacting production costs.
 - **How to Use:** Hover for monthly prices.
- **Diesel Prices Chart:**
 - A line chart comparing diesel prices in West and East Malaysia (in RM), used for machinery and logistics.
 - **How to Use:** Hover for price differences by date.

Navigation Tips

- Access via the "Commodity Trends" link in the sidebar (with a seedling icon).
- Charts are fixed at 96vh height for clarity; scroll if needed.
- Use these insights to assess external cost drivers for palm oil production.

Page 4: Geography Insights Tab

Overview

The Geography Insights tab provides geospatial data on palm oil estates and mills, with weather forecasts and environmental risk breakdowns.

Features

- **Palm Oil Map:**
 - Displays estates (circle markers) and mills (factory icons) in Malaysia.

- Estates are color-coded by weather: Green (Cerah), Yellow (Berangin), Orange (Hujan), Red (Ribut Petir).
- **How to Use:**
 - Use the **Date Slider** (bottom center) to view forecasts for different dates.
 - Hover over estates for details (e.g., plantation name, company, temperature).
 - Hover over mills for details (e.g., mill name, RSPO status).
 - Switch base maps (Streets, Satellite, Terrain) via the top-right layer control.
 - Toggle estates/mills visibility in the layer control.
- **7-Day Weather Forecast Summary:**
 - Shows counts of estates under each weather condition (Cerah, Berangin, Hujan, Ribut Petir) for a selected date.
 - Color-coded cards with icons: Green (Cerah), Yellow (Berangin), Orange (Hujan), Red (Ribut Petir).
 - **How to Use:** Select a date from the dropdown to update counts.
- **Palm Oil Related Environmental Risks:**
 - A stacked bar chart showing risk composition (Coastal Flood, Riverine Flood, Drought) for major companies, in percentages.
 - Derived from the Aqueduct Global Water Risk Atlas.
 - **How to Use:**
 - Use the dropdown to switch between risk types.
 - Hover for percentage contributions per company.
 - Read the description below for context (e.g., "Coastal floods occur when...").

Navigation Tips

- Access via the "Geography Insights" link in the sidebar (with a map-marker icon).
- The map (80vh height) may take a moment to load due to geospatial data.
- Zoom and pan the map to explore specific regions.

Page 5: Global Trade Chain Tab

Overview

The Global Trade Chain tab visualizes palm oil trade networks and Malaysia's export/import trends for palm oil end products.

Features

- **Global Palm Oil Trade Network:**
 - A network graph with countries as nodes and trade flows as arrows.
 - Node colors: Beige (Export Only), Green (Import Only), Cream (Both).
 - Node size reflects trade connections; arrow thickness shows trade value (FOB, in USD).
 - **How to Use:**
 - Use the **Year Slider** to select a year.
 - Click **Play** (▶) to animate through years; click again to pause.

- Select a country from the dropdown to highlight its trade flows.
- Hover over arrows for trade values (e.g., "USD 1.2M").
- View stats (right panel): Total Countries, Change in Countries, Total FOB Value, FOB Value Change (%).

- **Malaysia's Palm Oil End Products Export & Import Charts:**

- Two bar charts showing net trade (exports minus imports) for:
 - Animal, Vegetable Oils, Fats, Waxes.
 - Chemical and Related Products.
- Includes line overlays for exports and imports (in RM).
- **How to Use:**
 - Use the dropdown to switch between charts.
 - Hover for specific values (e.g., "Exports: RM 500M").

Navigation Tips

- Access via the "Global Trade Chain" link in the sidebar (with a shipping-fast icon).
- The trade network graph (600px height) is interactive; drag nodes if needed.
- Use the legend (right panel) to interpret node colors.

General Tips for Using Bursa PalmAI

- **Sidebar Navigation:** Click links in the left sidebar to switch tabs. The sidebar remains fixed for easy access.
- **Interactivity:** Most visuals are interactive—hover for details, use sliders/dropdowns to filter data.
- **Data Updates:** Bursa PalmAI fetches real-time data; delays may occur due to network issues. Refresh if visuals don't load.
- **Responsive Design:** The app adapts to desktops, tablets, and mobiles, but a larger screen enhances chart visibility.
- **Support:** For issues, contact the team via the About Us tab or the website.
- **Disclaimer:** Data and forecasts are for informational purposes only, not investment advice.

Enjoy exploring the palm oil industry with Bursa PalmAI's powerful, AI-driven insights!